

WD Blue SN580 1 TB PCIe Gen 4.0 NVMe SSD

An average DRAM-less drive

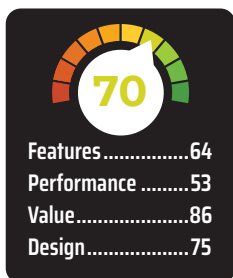
₹6,000



WD has built quite an impressive portfolio of SSDs over the years. Their WD_Black line up is known for being the premium tier with the Blue SSDs bringing in the mid-tier. WD has NVMe as well as SATA SSDs under their Blue lineup but this review is pertaining to the WD Blue SN580 NVMe SSD. It follows the insanely popular SN570 but in 2023-2024, the SN580 is poised as a mass market drive. There's one thing that might irk enthusiasts, and that's the lack of a DRAM chip but does it really matter? We find out.

SPECS AND FEATURES

The WD Blue SN580 is available in the usual set of capacities starting off at 250 GB and scaling all the way to 2 TB. It's a PCIe 4.0 NVMe drive with x4 PCIe lanes so it can make the most of the M.2 interface. WD has kept mum about their controllers for several generations now starting from the first NVMe SSD that they announced a couple of years ago. So there's not much we can tell you about the controller aside from the fact that it is designed and manufactured by SanDisk which is a subsidiary of Western Digital. Time and time again,



the proprietary SanDisk controllers have been proven to be very competitive in the storage space. The controller on the SN580 is the Polaris MP16+ controller which is ARM based like all controllers in the market and can support up to four flash chips at 1600

MT/s. That's about everything we know of this controller. The previous gen SN570 also used the same controller.

Moving onto the flash memory, the WD SN580 uses 112-layer Kioxia TLC (BiCS5) NAND chips. These chips were announced back in January 2020 and aren't the latest from Kioxia. In fact, Kioxia has already announced BiCS8 which is a 218-layer TLC NAND but we're yet to see products featuring the new NAND in the market. There's also BiCS9 which is lurking right around the corner but it will be some time before it becomes affordable to be put on consumer grade SSDs. The SN580 uses a single NAND design so the controller's full potential is not being leveraged. PCIe Gen 4.0 peak bandwidth is about 8 GB/s and the SN580 hits a little over half of that. Honestly speaking, there isn't much reason to go higher since WD has other PCIe 4.0 SSDs with higher speeds and they come with a heavier price tag. WD Blue was always about

offering good performance without breaking the bank. And using a single chip makes the manufacturing costs lower. Another point to note is that the previous gen SN570 also had the same BiCS5 NAND.

So, the WD SN580 has the same controller and the same NAND as the WD SN570. Which begs the question, why bother releasing a new model? Minor design changes aside, most of the changes will lie in the firmware. That doesn't mean that you should try flashing the firmware of the SN580 onto the SN570. You might mess things up if you can ever manage to flash it.

Speeds start off at 4000 MB/s sequential read on the 250 GB SKU to 4150 MB/s on the 2 TB SKU. And write speeds go from 2000 MB/s to 4150 MB/s as storage capacity goes up. One of the things that's on the lower side is the Endurance which is only 900 TBW for the 2 TB drive. There are several competing drives offering about 1200 TBW for their PCIe Gen 4.0 2 TB SKUs. The remaining SKUs seem to be at par with its contemporaries.

BUILD QUALITY

Like we mentioned, the SN580 utilises a single-NAND design. Also, it is a DRAM-less design since the controller comes with HMB (Host Memory Buffer) enabled. This means that a portion of the NAND flash is used by the controller as a pseudo-DRAM to boost performance. Aside from the controller and the NAND, there's also a Power Management Integrated Circuit (PMIC) on the unit. Overall, it uses a very simple design.

There's no heatsink on the SN580 because there's barely any heat generated. There is just the one NAND and the controller is not running anywhere close to its full potential so that doesn't put out any heat either. As a result, there's no aluminium heatsink and even the label sticker doesn't have any copper backing.

PERFORMANCE

Having the same controller and NAND